

AN EXACT HISTOGRAM FOR A QUADRATIC STAIRCASE

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ABSTRACT. For odd $n \geq 3$ let

$$W_j = \left\lfloor \frac{2(j+1)^2}{n} \right\rfloor - \left\lfloor \frac{2j^2}{n} \right\rfloor, \quad j = 0, 1, \dots, n-1.$$

We prove that $W_j \in \{0, 1, 2, 3, 4\}$ for every odd n and every j in this range, and that the five values occur with multiplicities determined by the single integer $A = \lfloor (n+7)/8 \rfloor$:

$$\#\{W=0\} = \#\{W=4\} = A, \quad \#\{W=2\} = 2A - 1, \quad \#\{W=1\} = \#\{W=3\} = \frac{n+1}{2} - 2A.$$

The proof is a tiling argument and the multiplicities are exact, with no error term. A probabilistic model of the same count, assuming $2j^2 \bmod n$ equidistributed, returns the main term $n/8$ and stays within 1 of the true value for every odd n , but it does not return that value. The exactness is a property of the tiling rather than of any averaging, and the residues $2j^2 \bmod n$ never enter the argument.

A companion compression is proved for the symmetric pairs (Theorems 3 and 4): a pair collapses to a single element of $\{0, 1, 2\}$, and its census is again governed by one integer.

We then prove the corresponding statement about the residues themselves, which the histogram deliberately avoids: the sequence $2j^2 \bmod n$ has exactly $2A$ interior local maxima for every odd n except $n = 3, 5, 7, 9, 49$ (Theorem 5). The argument is uniform for $n \geq 51$, and the twenty-four odd values $3 \leq n \leq 49$ are settled by direct computation, so the exception list belongs to the statement rather than being an observation about it. The proof combines a palindrome symmetry, a transition count that replaces the two conditions defining a maximum by one, the observation that the relevant *pairs* of intervals tile once separated by parity, and a small arithmetic coincidence: the four numbers $8x \bmod n$ that arise have signed representatives $\pm 1, \pm 3, \pm 5, \pm 7$ in some order, so the squares of those representatives always sum to 84.

Nothing in this paper concerns prime numbers.

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1. THE OBJECT

Throughout, $n \geq 3$ is odd and j runs over $0, 1, \dots, n-1$. Set

$$F(x) = \left\lfloor \frac{2x^2}{n} \right\rfloor, \quad W_j = F(j+1) - F(j), \quad r_j = 2j^2 \bmod n.$$

F is a staircase under the parabola $2x^2/n$; W_j is its increment, and r_j is the residue left over. The two are linked by $2j^2 = nF(j) + r_j$, but — and this is the point of Section 3 — the histogram of W can be determined without ever computing an r_j .

Two elementary remarks fix the scale. The total rise is

$$\sum_{j=0}^{n-1} W_j = F(n) - F(0) = 2n,$$

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so the mean increment is exactly 2; and W_j counts the multiples of n in the interval

$$I_j = (2j^2, 2(j+1)^2], \quad |I_j| = 4j + 2. \quad (1.1)$$

The intervals $I_0, I_1, \dots$ tile $(0, \infty)$. That is the only structural fact the histogram needs.

2. THE RANGE

Theorem 1. *For every odd n and every $0 \leq j \leq n-1$,*

$$W_j \in \{0, 1, 2, 3, 4\}.$$

Proof. The two arguments of the floor differ by

$$\frac{2(j+1)^2}{n} - \frac{2j^2}{n} = \frac{4j+2}{n},$$

which is positive and, for $j \leq n-1$, strictly less than 4. A difference of floors of two reals differing by less than 4 lies in $\{0, 1, 2, 3, 4\}$. $\square$

Verification. Zero violations over every odd $n \leq 20,001$.

The restriction $j \leq n-1$ is real: for j in the range $[rn, (r+1)n)$ the same computation gives $W_j \in \{4r, \dots, 4r+4\}$, so the bound moves up in steps of four. Everything below concerns $0 \leq j \leq n-1$, which we call the *first cycle*.

Two further symmetries, both immediate from (1.1), will be used:

$$W_j + W_{n-1-j} = 4, \quad W_{(n-1)/2} = 2. \quad (2.1)$$

3. THE EXACT MULTIPLICITIES

Theorem 2. *Let $n \geq 3$ be odd and $A = \lfloor (n+7)/8 \rfloor$. Over the first cycle,*

$$\#\{W=0\} = \#\{W=4\} = A, \quad \#\{W=2\} = 2A - 1, \quad \#\{W=1\} = \#\{W=3\} = \frac{n+1}{2} - 2A,$$

and no other value occurs.

n	(#0, #1, #2, #3, #4)	A
11	(2, 2, 3, 2, 2)	2
101	(13, 25, 25, 25, 13)	13
1,009	(127, 251, 253, 251, 127)	127
19,997	(2500, 4999, 4999, 4999, 2500)	2500

Verification. Zero exceptions for every odd $n \leq 200,001$.

Proof. By (1.1), W_j counts the multiples of n in I_j , an interval of length exactly $4j+2$. Hence:

- if $4j+2 < n$, the interval is shorter than n and holds *at most one* multiple, so $W_j \in \{0, 1\}$;
- if $4j+2 \geq n$, it is at least as long and holds *at least one*, so $W_j \geq 1$.

The count of zeros. Consequently $W_j = 0$ forces $j < m := \lfloor (n+1)/4 \rfloor$. On that range the intervals $I_0, \dots, I_{m-1}$ tile $(0, 2m^2]$ exactly and each holds at most one multiple, so the number of *occupied* intervals equals the number of multiples in the range, namely $\lfloor 2m^2/n \rfloor$. Therefore

$$N_0 = m - \left\lfloor \frac{2m^2}{n} \right\rfloor. \quad (3.1)$$

r	m	q	$2m^2 - nq$	check
1	$2t$	$t - 1$	$7t + 1$	$< 8t + 1$
3	$2t + 1$	t	$5t + 2$	$< 8t + 3$
5	$2t + 1$	t	$3t + 2$	$< 8t + 5$
7	$2t + 2$	$t + 1$	$t + 1$	$< 8t + 7$

Write $n = 8t + r$ with $r \in \{1, 3, 5, 7\}$; then $m = 2t + \varepsilon$ with $\varepsilon = 0, 1, 1, 2$ respectively. We claim the quotient in (3.1) is $q = t + \varepsilon - 1$:

Each remainder is a positive linear function of t strictly below n , so the quotient is as claimed and

$$N_0 = (2t + \varepsilon) - (t + \varepsilon - 1) = t + 1 = \left\lfloor \frac{n+7}{8} \right\rfloor = A. \quad (3.2)$$

The remaining four. Put $H = (n-1)/2$. For $0 \leq j < H$ the interval I_j has length $4j+2 < 2n$, so $W_j \in \{0, 1, 2\}$. These H intervals tile $(0, 2H^2]$, which contains

$$\left\lfloor \frac{2H^2}{n} \right\rfloor = H - 1$$

multiples of n . Writing $N_i^- = \#\{0 \leq j < H : W_j = i\}$, we have $N_0^- = A$ together with

$$N_0^- + N_1^- + N_2^- = H, \quad N_1^- + 2N_2^- = H - 1,$$

and subtracting gives $N_2^- = A - 1$. The central index $j = H$ has $W_H = 2$ by (2.1). Finally the symmetry $W_j + W_{n-1-j} = 4$ pairs $W=0$ with $W=4$ and $W=1$ with $W=3$, while $W=2$ pairs with itself. Therefore

$$N_4 = N_0 = A, \quad N_2 = 2(A - 1) + 1 = 2A - 1,$$

and the remaining $n - (N_0 + N_2 + N_4)$ indices split equally:

$$N_1 = N_3 = \frac{n+1}{2} - 2A. \quad \square$$

3.1. Why the count has no error term. A probabilistic model, assuming $2j^2 \bmod n$ equidistributed, estimates N_0 by

$$\sum_{j < (n-2)/4} \frac{n - 4j - 2}{n} = \frac{n^2 - 1}{8n},$$

the sum being exactly that for every odd n , since it telescopes to $m - 2m^2/n$ with $m = \lfloor (n+1)/4 \rfloor$. The true value is $N_0 = A = \lfloor n/8 \rfloor$, so the model returns the main term $n/8$ and not the value itself: the difference $A - (n^2 - 1)/(8n)$ lies in $[0, 1)$ for every odd n and reaches 0.889 at $n = 9$, where the model gives 10/9 against a true count of 2.

The exactness of Theorem 2 is a property of the tiling and not of any averaging. The intervals I_j partition the range, so the number of occupied ones is a difference of two floors taken at the two ends, and nothing is estimated anywhere in the argument. The residues $2j^2 \bmod n$ never enter it. Theorem 5 is the statement about those residues, and it is a good deal harder.

4. PAIRS

Index the symmetric pairs $\{j, n-1-j\}$ by $R = 1, 3, \dots, n-2$.

Theorem 3. *With*

$$d_n(R) = \left\lfloor \frac{1}{2} + \frac{(R+2)^2}{2n} \right\rfloor - \left\lfloor \frac{1}{2} + \frac{R^2}{2n} \right\rfloor,$$

one has $d_n(R) \in \{0, 1, 2\}$, and the corresponding pair of increments is $(W_L, W_R) = (2 - d, 2 + d)$.

Proof. The two arguments of the floor differ by

$$\frac{(R+2)^2 - R^2}{2n} = \frac{2(R+1)}{n},$$

which for $1 \leq R \leq n-2$ lies strictly between 0 and 2; hence the difference of floors is 0, 1 or 2. Substituting the pair parametrisation into the definition of W gives $W_R = 2 + d_n(R)$, and (2.1) then gives $W_L = 2 - d_n(R)$. $\square$

Verification. Zero failures over every odd $n < 600$ and every odd R with $1 \leq R \leq n-2$.

Theorem 4. *Let $P_d = \#\{R : d_n(R) = d\}$, written P rather than N to keep it apart from the multiplicities of Theorem 2. Then $P_2 = P_0 + 1$.*

Proof. There are $(n-1)/2$ odd values $R = 1, 3, \dots, n-2$, and the definition of $d_n(R)$ telescopes along them:

$$\sum_R d_n(R) = \left\lfloor \frac{1}{2} + \frac{n^2}{2n} \right\rfloor - \left\lfloor \frac{1}{2} + \frac{1}{2n} \right\rfloor = \frac{n+1}{2}.$$

Hence

$$P_0 + P_1 + P_2 = \frac{n-1}{2}, \quad P_1 + 2P_2 = \frac{n+1}{2},$$

and subtracting gives $P_2 - P_0 = 1$. $\square$

Verification. Zero failures over all odd $n < 2000$.

n	N_0	N_1	N_2
11	1	2	2
13	1	3	2
31	3	8	4
101	12	25	13
499	62	124	63

As in Theorem 2, the whole census is governed by one integer.

5. THE NUMBER OF LOCAL MAXIMA OF $2j^2 \bmod n$

Theorem 2 counts the values of W . The companion question concerns their *arrangement*, and its sharpest form is the number of local maxima of the residue sequence itself.

Theorem 5. *Let $n \geq 3$ be odd and $A = \lfloor (n+7)/8 \rfloor$. Then $r_j = 2j^2 \bmod n$, $j = 0, \dots, n-1$, has exactly $2A$ interior local maxima, except at the five values $n = 3, 5, 7, 9, 49$.*

The maximum is meant strictly: j is counted when $r_j > r_{j-1}$ and $r_j > r_{j+1}$. The proof below reads a maximum through the weaker pair of conditions $\varepsilon_{j-1} = 0$ and $\varepsilon_j = 1$, and the two readings agree from $n = 11$ on.

Lemma (the two readings of a maximum). *For odd $n \geq 11$ and $1 \leq j \leq n-2$, the conditions $\varepsilon_{j-1} = 0$ and $\varepsilon_j = 1$ hold together if and only if j is a strict local maximum.*

Proof. The two readings can differ only where $r_j = r_{j-1}$. That forces $2j^2 \equiv 2(j-1)^2$, i.e. $n \mid 2(2j-1)$, and n is odd, so $n \mid 2j-1$; for $1 \leq j \leq n-1$ the only solution is $j = H+1$ with $H = (n-1)/2$. Doubling identifies the two residues there: $2r_H \equiv (n-1)^2 \equiv 1$ and $2 \cdot \frac{n+1}{2} = n+1 \equiv 1$, so $r_H = r_{H+1} = (n+1)/2$; and $2r_{H-1} \equiv (n-3)^2 \equiv 9$ with $2 \cdot \frac{n+9}{2} = n+9 \equiv 9$, so $r_{H-1} = (n+9)/2$ as soon as $(n+9)/2 < n$, that is $n > 9$. Since $r_{H+2} = r_{n-H-2} = r_{H-1}$, we get $r_{H+2} > r_{H+1}$, so $\varepsilon_{H+1} = 0$ and the weaker reading does not fire at $j = H+1$ either. $\square$

At $n = 5, 7, 9$ the quantity $(n + 9)/2$ is not below n , the last step fails, and the two readings do differ; all three are among the five exceptions in any case.

Range of the statement. The argument below is uniform for $n \geq 51$. Two things need it: the interval structure of Step 4 is in its generic configuration once $n \geq 13$, and the Lemma of Step 5 needs $e_x^2 < n$, which holds exactly when $n > 49$. The twenty-four odd values $3 \leq n \leq 49$ are settled instead by direct computation, and five of them fail — $n = 3, 5, 7, 9, 49$ — so the statement above, exception list included, is proved rather than observed. The count was checked against the direct definition for all odd $n \leq 200,001$ as well.

Throughout put

$$\varepsilon_j = [r_{j+1} < r_j], \quad q_j = \left\lfloor \frac{4j+2}{n} \right\rfloor, \quad \Phi(x) = \left\lfloor \frac{2x^2}{n} \right\rfloor + \left\lfloor \frac{2(x-1)^2}{n} \right\rfloor, \quad H = \frac{n-1}{2},$$

so that $\varepsilon_j = W_j - q_j$, and j is a local maximum exactly when $\varepsilon_{j-1} = 0$ and $\varepsilon_j = 1$.

5.1. Step 1: the palindrome. Since $2(n-j)^2 \equiv 2j^2$, we have $r_j = r_{n-j}$ for $1 \leq j \leq n-1$. Maxima therefore occur in mirror pairs $j \leftrightarrow n-j$, with no fixed point because n is odd. The pairing stays inside the interior indices because $j = 1$ carries no maximum once $n \geq 9$, where $r_1 = 2$ and $r_2 = 8$; at $n = 7$ it does carry one, and its mirror $n-1$ is not an interior index, which is one of the reasons 7 is an exception. Also $r_{H+1} = r_{n-H-1} = r_H$, so

$$\varepsilon_0 = 0 \quad (\text{since } r_1 = 2 > 0 = r_0), \quad \varepsilon_H = 0,$$

and no maximum straddles the middle. Hence the total is twice the number of maxima in $1 \leq j \leq H$.

5.2. Step 2: maxima are exactly the mixed pairs. In any 0–1 word the numbers of ascents and descents differ by the boundary values; applied to $\varepsilon_0, \dots, \varepsilon_H$ with both ends 0, the two are equal. Therefore

$$\#\{j \leq H : \varepsilon_{j-1} = 0, \varepsilon_j = 1\} = \frac{1}{2} \#\{j \leq H : \varepsilon_{j-1} \neq \varepsilon_j\},$$

and combining with Step 1,

$$\#\{\text{local maxima}\} = \#\{1 \leq j \leq H : \varepsilon_{j-1} \neq \varepsilon_j\}.$$

5.3. Step 3: a mixed pair is one condition on one interval. Since $\varepsilon_{j-1} + \varepsilon_j = (W_{j-1} + W_j) - (q_{j-1} + q_j)$, the pair at j is mixed if and only if

$$P_j := (2(j-1)^2, 2(j+1)^2] \quad \text{contains exactly } q_{j-1} + q_j + 1 \text{ multiples of } n.$$

P_j has length exactly $8j$. No residue occurs in this condition.

5.4. Step 4: the pair intervals tile, by parity. Consecutive P_j overlap, but each parity subfamily tiles exactly:

$$P_{2s+1} = (8s^2, 8(s+1)^2], \quad P_{2s} = (2(2s-1)^2, 2(2s+1)^2],$$

of lengths $8(2s+1)$ and $16s$. Since $8j < 4n$ on $j \leq H$, each P_j carries L_j or $L_j + 1$ multiples with $L_j = \lfloor 8j/n \rfloor \in \{0, 1, 2, 3\}$, and over any block of consecutive j of one parity the total telescopes. Set $\text{off}_j = (q_{j-1} + q_j) - L_j$; then j is mixed if and only if P_j carries the *larger* count when $\text{off}_j = 0$, and the *smaller* when $\text{off}_j = -1$.

Both L_j and off_j are explicit step functions of j , with jumps only at

$$1 < \ell_1 = \left\lceil \frac{n}{8} \right\rceil \leq \beta_1 = \left\lceil \frac{n-2}{4} \right\rceil \leq \ell_2 = \left\lceil \frac{n}{4} \right\rceil \leq \beta_2 = \left\lceil \frac{n+2}{4} \right\rceil < \ell_3 = \left\lceil \frac{3n}{8} \right\rceil < H < H + 1,$$

giving *seven* intervals with $(L, \text{off}) = (0, 0), (1, -1), (1, 0), (2, -1), (2, 0), (3, -1), (3, 0)$ in order. The last is the single point $j = H$, where $4H + 2 = 2n$ forces $q_H = 2$. Combining the two parities,

an interval $[\alpha, \beta)$ has $\beta - \alpha$ tiles whose multiples total $\Phi(\beta) - \Phi(\alpha)$, independently of which parity starts it. Its contribution to the mixed count is $G - LC$ if $\text{off} = 0$ and $(L + 1)C - G$ if $\text{off} = -1$, with $G = \Phi(\beta) - \Phi(\alpha)$ and $C = \beta - \alpha$.

Summing the seven intervals and telescoping the Φ 's (note $\Phi(1) = 0$):

$$\# \text{mixed} = 2[\Phi(\ell_1) - \Phi(\beta_1) + \Phi(\ell_2) - \Phi(\beta_2) + \Phi(\ell_3) - \Phi(H)] + \Phi(H + 1) + \Sigma_C, \quad (5.1)$$

$$\Sigma_C = 2(\beta_1 - \ell_1) - (\ell_2 - \beta_1) + 3(\beta_2 - \ell_2) - 2(\ell_3 - \beta_2) + 4(H - \ell_3) - 3.$$

Verification. Equation (5.1): zero failures against the direct count for every odd $9 \leq n \leq 20,001$.

5.5. Step 5: evaluating the seven Φ 's. Two are immediate: $\Phi(H+1) = n-1$ and $\Phi(H) = n-5$. Three more are linear; writing $n = 8t + r$ with $r \in \{1, 3, 5, 7\}$,

$$-\Phi(\beta_1) + \Phi(\ell_2) - \Phi(\beta_2) = -2t + c_r, \quad c_r = 3, -1, 1, -3.$$

The remaining two are the content, and they are handled together.

Lemma. For odd $n = 8t + r \geq 51$,

$$\Phi(\lceil n/8 \rceil) + \Phi(\lceil 3n/8 \rceil) = \frac{5n - k_r}{8}, \quad k_r = 13, 7, 25, 19 \text{ for } r = 1, 3, 5, 7.$$

Proof. Put $m = \lceil 3r/8 \rceil \in \{1, 2, 2, 3\}$, so $\lceil n/8 \rceil = t + 1$ and $\lceil 3n/8 \rceil = 3t + m$; the four arguments are $x \in \{t, t + 1, 3t + m - 1, 3t + m\}$. Because $8t \equiv -r \pmod{n}$,

$$8x \equiv e_x \pmod{n}, \quad e_x = -r, 8 - r, 8m - 8 - 3r, 8m - 3r,$$

and in every one of the four cases $\{|e_x|\}$ is a permutation of $\{1, 3, 5, 7\}$ — for instance $r = 1, m = 1$ gives $(-1, 7, -3, 5)$ and $r = 7, m = 3$ gives $(-7, 1, -5, 3)$. Since $32 \cdot 2x^2 = (8x)^2$,

$$32(2x^2 \bmod n) \equiv e_x^2 \pmod{n}, \quad \sum_x e_x^2 = 1 + 9 + 25 + 49 = 84.$$

Write $32(2x^2 \bmod n) = e_x^2 + a_x n$. That a_x is an integer in $[1, 32)$ is the one place the hypothesis $n \geq 51$ is used: it gives $e_x^2 \leq 49 < n$, so e_x^2 is already the reduced residue of the left side and $a_x \geq 0$, while $a_x = 0$ would force $32 \mid e_x^2$, false for $e_x^2 \in \{1, 9, 25, 49\}$; the upper bound is automatic from $2x^2 \bmod n < n$. So a_x is the unique such integer with $e_x^2 + a_x n \equiv 0 \pmod{32}$, that is $a_x \equiv -n^{-1}e_x^2 \pmod{32}$. Now $1^2, 3^2, 5^2, 7^2$ are $1, 9, 25, 17$ modulo 32, which are precisely the four classes there that are $\equiv 1 \pmod{8}$; multiplying by the odd number $-n^{-1}$ sends $1 + 8k$ to $-n^{-1} + 8k'$ with k' running over $0, 1, 2, 3$ again, so the four values a_x are $c, c + 8, c + 16, c + 24$ for the single $c \in [1, 8]$ with $c \equiv -n^{-1} \pmod{8}$. Since $n \equiv r$ and $r^2 \equiv 1 \pmod{8}$, we have $c = 8 - r$, and therefore $\sum_x a_x = 4c + 48 = 80 - 4r$, whence

$$\sum_x (2x^2 \bmod n) = \frac{84 + (80 - 4r)n}{32} = \frac{(20 - r)n + 21}{8}.$$

On the other hand $\sum_x 2x^2 = 40t^2 + (24m - 8)t + (4m^2 - 4m + 4)$ exactly, and dividing the difference by $n = 8t + r$ gives $5t + s_r$ with $s_r = -1, 1, 0, 2$, which is $(5n - k_r)/8$. $\square$

Verification. The Lemma: zero failures for every odd $n \leq 300,001$ apart from $n = 3, 5, 7, 9, 17, 25, 49$. At each of those, $e_x^2 < n$ fails for one of the four x : that a_x comes out 32 below the value the congruence alone would give, the sum $\sum_x (2x^2 \bmod n)$ is short by exactly n , and the Lemma's stated value is 1 below the truth.

n	$8t + 1$	$8t + 3$	$8t + 5$	$8t + 7$
$\#\text{mixed}$	$2t + 2$	$2t + 2$	$2t + 2$	$2t + 2$

5.6. Step 6: assembly. Substituting $\ell_1 = t + 1$, $\beta_1 = 2t + \lceil \frac{r-2}{4} \rceil$, $\ell_2 = 2t + \lceil \frac{r}{4} \rceil$, $\beta_2 = 2t + \lceil \frac{r+2}{4} \rceil$, $\ell_3 = 3t + m$, $H = 4t + \frac{r-1}{2}$ into (5.1) together with the Lemma gives, in each of the four residue classes,

$$\#\text{mixed} = 2t + 2 = 2A. \quad \blacksquare$$

Where the small values break, and why the two exception lists differ. Identity (5.1) itself holds for every odd $n \geq 9$, $n = 49$ included, so nothing goes wrong in Step 4 there; what fails below 51 is the evaluation in Step 5. At $n = 17$ and $n = 25$ the true value of $\Phi(\lceil n/8 \rceil) + \Phi(\lceil 3n/8 \rceil)$ exceeds the Lemma by 1 while the true linear part falls short of $-2t + c_r$ by 1, the two cancel in the bracket, and the conclusion survives. At $n = 49$ only the Lemma fails, the bracket is short by 1, and the count comes out $2A + 2 = 16$ against the predicted 14. At $n \leq 9$ several of the evaluations fail at once. That is why the exception list of Theorem 5 and the exception list of the Lemma are not the same list.

5.7. Remark: what made it work. A local maximum is *two* consecutive conditions, while Theorem 2 counts j satisfying *one*; and while the intervals I_j tile, consecutive P_j overlap. The apparent obstruction dissolves in two moves. Steps 2 and 3 convert the two conditions into one — a condition on the count of multiples in the single interval P_j — and Step 4 observes that the P_j tile perfectly once separated by parity. The elementary weight $\sum_j \varepsilon_j = (n-1)/2$, which follows from $\gcd(4, n) = 1$ because the increments $(4j+2) \bmod n$ then run over every residue, is thereby joined by a second weight, of pairs rather than of single terms, and the same tiling carries both.

The one place where genuine arithmetic enters is the Lemma, and it enters through a small coincidence worth naming: the four numbers $8x \bmod n$ attached to $\lceil n/8 \rceil$ and $\lceil 3n/8 \rceil$ have signed representatives $e_x = \pm 1, \pm 3, \pm 5, \pm 7$ in some order, so $\sum_x e_x^2 = 84$ whatever n is. It is those representatives, not the reduced residues themselves, whose squares sum to 84.

6. PLACEMENT

Since $r_j = 2j^2 \bmod n$ is $n\{j^2\alpha\}$ with $\alpha = 2/n$ rational, the natural comparisons are these.

The linear analogue, and what it names. Replace j^2 by j . The difference sequence $u_j = \lfloor (j+1)\alpha \rfloor - \lfloor j\alpha \rfloor$ then takes values in $\{0, 1\}$ and is the characteristic *Sturmian* word of slope α ; the exact frequencies of its factors are classical, and they are read off from the three-distance theorem of Sós, Świerczkowski and Surányi [5] — see Alessandri and Berthé [1] for that connection made explicit. In these terms Theorem 1 says that W_j is obtained from the same expression with j replaced by j^2 , over the alphabet $\{0, 1, 2, 3, 4\}$ instead of $\{0, 1\}$, and Theorem 2 is its letter census.

The analogy is one of form only, and it is worth saying how far it fails. A Sturmian word is defined by its properties — a two-letter alphabet, factor complexity $p(m) = m + 1$ for every m , aperiodicity, and imbalance at most 1 — and not by the displayed formula, which merely characterises it in the irrational linear case. The word W has none of those properties. It is periodic; its alphabet has five letters; and its factor complexity, measured on the first cycle, stabilises for large n at

$$p(1), \dots, p(8) = 5, 14, 31, 56, 97, 146, 219, 308$$

(identical at $n = 4001, 10007, 20011$), against the Sturmian 2, 3, 4, 5, 6, 7, 8, 9; its maximum imbalance grows with n , measured at 3, 25, 39 for $n = 11, 101, 1009$, against the Sturmian value

1. We found no closed form for the complexity. The point of the comparison is therefore the name of the object and nothing more — and it is doubly a comparison rather than a specialisation, since the linear word at a *rational* slope is periodic and so is not Sturmian either.

The comparison is worth making because of the one place the two cases part. For *rational* α the linear word is periodic and its letter frequencies are immediate. Here the slope is the rational $2/n$, the word is likewise periodic — and the multiplicities are not immediate at all: they are the content of Theorem 2.

Differences of floor functions are of course studied in their own right, and it is worth saying precisely how the nearest such work differs from the present object. O'Bryant [4] obtains generating functions for sequences $\lfloor an/x \rfloor - \lfloor bn/x \rfloor$ and, from them, Beatty-type partition theorems for $\lfloor 2nx \rfloor - \lfloor nx \rfloor$. That difference is taken between two dilates of one *linear* sequence at a common index, and the method requires x irrational, since it runs over the discontinuities of a step function. Ours is the first difference *along the index* of a *quadratic* staircase at a *rational* parameter, and what is counted is the multiplicity of each letter rather than a partition of the integers. We have not located the quadratic case there or elsewhere, and we state the difference rather than the resemblance because it is the difference that leaves the question open. Likewise we do not claim the three-distance theorem as a template: what the linear case supplies is the name of the family, not the argument.

The quadratic literature runs the other way. For irrational α the fine-scale statistics of $\{j^2\alpha\}$ — pair correlation and spacings — are studied as limiting laws over a *continuum* of α . Heath-Brown [2] is quantitative as well as metric: he constructs a dense set of α for which the pair correlation is Poissonian, and for Diophantine α he bounds the pair correlation function by X with an error of size $X^{7/8}$ in the range $1 \leq X \leq \log N$ — a bound at a fixed scale rather than a limit. A statement of the present kind — a finite set of values with explicit multiplicities and no error term — uses the fact that $\alpha = 2/n$ is rational here, so that the orbit is finite and periodic. We use that structure; we do not claim it is the only route to such a statement.

The nearest relative is the literature on spacings of quadratic residues modulo q , for instance Kurlberg [3]. The results there are again distributional; Theorems 2, 4 and 5 are exact identities for a single explicit sequence.

We are not aware of the multiplicities of Theorem 2, the pair census of Theorem 4, or the count of Theorem 5 in the literature, and would be glad to be corrected.

APPENDIX A. SAMPLE DATA

The increments W_j , $n = 7$: 0, 1, 1, 2, 3, 3, 4. Here $A = \lfloor 14/8 \rfloor = 1$, and Theorem 2 predicts $\#\{0\} = \#\{4\} = 1$, $\#\{2\} = 2A - 1 = 1$, $\#\{1\} = \#\{3\} = (n+1)/2 - 2A = 2$ — exactly what the list shows. (Note that $n = 7$ is one of the five small exceptions to Theorem 5; the histogram of Theorem 2 has no exceptions at all.)

The residues $r_j = 2j^2 \bmod 17$: 0, 2, 8, 1, 15, 16, 4, 13, 9, 9, 13, 4, 16, 15, 1, 8, 2. Interior local maxima at $j = 2, 5, 7, 10, 12, 15$ — six of them, and $2A = 2 \lfloor 24/8 \rfloor = 6$.

The seven intervals of Step 4, $n = 101$ ($t = 12$, $r = 5$, $H = 50$): the boundaries are

$$1, \quad \ell_1 = 13, \quad \beta_1 = 25, \quad \ell_2 = 26, \quad \beta_2 = 26, \quad \ell_3 = 38, \quad H = 50, \quad H + 1 = 51,$$

so the fourth interval $[\ell_2, \beta_2)$ is empty here and six of the seven carry indices, with $(L, \text{off}) = (0, 0)$ on $[1, 13)$, $(1, -1)$ on $[13, 25)$, $(1, 0)$ at $j = 25$, $(2, 0)$ on $[26, 38)$, $(3, -1)$ on $[38, 50)$ and $(3, 0)$ at $j = 50$.

APPENDIX B. NOTE ON METHOD

Two details were found by failure rather than by design and are recorded so that a reader reconstructing the argument does not lose time on them.

The constant 2 in the increment. The author met this staircase first in a setting where the quantity of interest is $2 + W_j$ rather than W_j ; dropping the additive constant breaks the formula in 57 of 78 tested cases. The constant is an artefact of that setting and not of the staircase, but it is easy to mislay.

The seventh interval. A first version of (5.1) used six intervals and failed for essentially every n . The omission is the single point $j = H$, where $4H + 2 = 2n$ forces $q_H = 2$ rather than 1; with it restored the identity holds with zero failures on the whole tested range.

REPRODUCIBILITY

Every count reported above — the range of W_j , the five multiplicities, the pair census, the seven-interval identity (5.1), the Lemma of Step 5, and the number of local maxima — is regenerated by a single script that exits with a non-zero status if any check fails. It is available from the author on request.

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